

Gauss-Markov Framework for Linear Models

1 General Gauss-Markov model

Consider independent observations $Y_1, Y_2, \dots, Y_n$ such that

$$Y_i = x_{i1}\beta_1 + \dots + x_{ip}\beta_p + \varepsilon_i, \quad \mathbb{E}(\varepsilon_i) = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad i = 1, 2, \dots, n, \quad (1)$$

and ε_i 's are independent. Here $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$ and σ^2 are unknown parameters and $\mathbf{X} = ((x_{ij}))$ is an $n \times p$ matrix of known values of p predictors $X_1, \dots, X_p$. $\mathbf{X}$ can be observations from measured covariates (e.g. multiple regression) or may be a fixed “design matrix” (e.g. one way classified data, i.e., one way ANOVA) or both (e.g. ANCOVA model).

- If $\mathbf{Y}$ stands for the $n \times 1$ column vector $(Y_1, \dots, Y_n)^T$, we can write the model in matrix notation as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

- The model implies $\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$.
- One problem is to estimate linear functions of $\boldsymbol{\beta}$ and σ^2 on the basis of observations Y_i .
- Another problem is to estimate the noise variance σ^2 .

1.1 Least squares estimator

We now consider methods for estimating $\boldsymbol{\beta}$. Consider the objective function

$$G(\mathbf{b}) = (\mathbf{Y} - \mathbf{X}\mathbf{b})^T(\mathbf{Y} - \mathbf{X}\mathbf{b}) \quad \mathbf{b} \in \mathbb{R}^p.$$

A vector $\mathbf{b} = \hat{\boldsymbol{\beta}}$ that minimizes $G(\mathbf{b})$ is called a *least squares estimator* of $\boldsymbol{\beta}$. We summarize below a few basic properties.

- The minimum of $G(\mathbf{b})$ is attained at $\hat{\boldsymbol{\beta}}$ if and only if $\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{Y}$.
- The set of equations $\mathbf{X}^T \mathbf{X} \mathbf{b} = \mathbf{X}^T \mathbf{Y}$ is called the set of *normal equations*.
- If $\mathbf{X}$ has full column rank p , i.e., $\mathbf{X}^T \mathbf{X}$ is nonsingular, then the least squares estimator of $\boldsymbol{\beta}$ is unique and is given by $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$.
- In general, any solution $\hat{\boldsymbol{\beta}}$ of the normal equations is of the form $(\mathbf{X}^T \mathbf{X})^- \mathbf{X}^T \mathbf{Y} + \mathbf{z}$ where $(\mathbf{X}^T \mathbf{X})^-$ is a g-inverse of $\mathbf{X}^T \mathbf{X}$ and $\mathbf{z} \in \mathbb{R}^p$ is any vector such that $\mathbf{X}^T \mathbf{X} \mathbf{z} = \mathbf{0}$.
- The $n \times 1$ vector $\hat{\mathbf{Y}}$ of *fitted values*, denoted $\hat{Y}_i = \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_p x_{ip}$, for $i = 1, \dots, n$, can be expressed as

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^- \mathbf{X}^T \mathbf{Y} = \mathbf{P}_X \mathbf{Y}, \quad (2)$$

where $\mathbf{P}_X = \mathbf{X}(\mathbf{X}^T \mathbf{X})^- \mathbf{X}^T$ is the orthogonal projection into $\mathbf{X}$ = the column space of $\mathbf{X}$, where $(\mathbf{X}^T \mathbf{X})^-$ is any g-inverse of $\mathbf{X}^T \mathbf{X}$. Note that $\hat{\mathbf{Y}}$ is always unique even though $\hat{\boldsymbol{\beta}}$ may not be a unique solution of the normal equations.

- The *residual vector* $\mathbf{e} = (e_1, \dots, e_n)^T$, where $e_i = \hat{\varepsilon}_i = Y_i - \hat{Y}_i$ is then expressed as $\mathbf{e} = (\mathbf{I}_n - \mathbf{P}_X) \mathbf{Y}$. Observe that $\mathbf{X}^T \mathbf{e} = \mathbf{0}$ and hence $\hat{\mathbf{Y}}^T \mathbf{e} = 0$.

1.2 Estimation of error variance σ^2

Definition 1 The *residual sum of squares* (*RSS*) or *Sum of Squares of Errors* (*SSE*) under the general Gauss-Markov model is defined as

$$\begin{aligned} SSE &:= \sum_{i=1}^n e_i^2 = \|\mathbf{e}\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 \\ &= \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 \\ &= \|\mathbf{Y} - \mathbf{P}_X \mathbf{Y}\|^2 = \mathbf{Y}^T (\mathbf{I}_n - \mathbf{P}_X) \mathbf{Y}. \end{aligned} \quad (3)$$

The following result shows that an unbiased estimator of σ^2 is given by the *Mean Squared Error* (MSE), defined as $SSE/df(SSE)$, where $df(SSE)$ stands for the *degrees of freedom* of SSE, and the latter equals $n - r$, where $r = \text{rank}(\mathbf{X})$.

Proposition 1 If $\text{rank}(\mathbf{X}) = r$ with $1 \leq r \leq p < n$, then $\mathbb{E}(SSE) = (n - r)\sigma^2$.

Proof: We start by noting that, since $\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$,

$$\text{Var}(\mathbf{Y}) = \mathbb{E}[(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T] = \sigma^2 \mathbf{I}_n.$$

From this, we obtain

$$\mathbb{E}(\mathbf{Y}\mathbf{Y}^T) = \mathbf{X}\boldsymbol{\beta}\boldsymbol{\beta}^T \mathbf{X}^T + \sigma^2 \mathbf{I}_n. \quad (4)$$

From the representation of SSE , we have $\mathbb{E}(SSE) = \mathbb{E}(\mathbf{Y}^T (\mathbf{I}_n - \mathbf{P}_X) \mathbf{Y})$. Now, $\mathbf{Y}^T (\mathbf{I} - \mathbf{P}) \mathbf{Y}$ is a scalar, and so $\mathbf{Y}^T (\mathbf{I}_n - \mathbf{P}_X) \mathbf{Y} = \text{tr}(\mathbf{Y}^T (\mathbf{I}_n - \mathbf{P}_X) \mathbf{Y})$, where tr denotes trace. Also from the property of trace, we get

$$\text{tr}(\mathbf{Y}^T (\mathbf{I}_n - \mathbf{P}_X) \mathbf{Y}) = \text{tr}((\mathbf{I}_n - \mathbf{P}_X) \mathbf{Y} \mathbf{Y}^T).$$

Now, since trace is a linear operation, by *linearity of expectation*,

$$\begin{aligned} \mathbb{E}[\text{tr}((\mathbf{I}_n - \mathbf{P}_X) \mathbf{Y} \mathbf{Y}^T)] &= \text{tr}((\mathbf{I}_n - \mathbf{P}_X) \mathbb{E}(\mathbf{Y} \mathbf{Y}^T)) \\ &= \text{tr}((\mathbf{I}_n - \mathbf{P}_X) (\mathbf{X}\boldsymbol{\beta}\boldsymbol{\beta}^T \mathbf{X}^T + \sigma^2 \mathbf{I}_n)) \quad (\text{by (4)}) \\ &= \sigma^2 \text{tr}(\mathbf{I}_n - \mathbf{P}_X), \quad (\text{since } \mathbf{P}_X \mathbf{X} = \mathbf{X}) \\ &= \sigma^2 (n - \text{tr}(\mathbf{P}_X)) = \sigma^2 (n - r), \end{aligned}$$

where in the last step, we used the fact that $\mathbf{P}_X$ is an idempotent matrix, so that $\text{tr}(\mathbf{P}_X) = \text{rank}(\mathbf{P}_X) = \text{rank}(\mathbf{X})$.

1.3 Estimation of linear functions of $\boldsymbol{\beta}$

Definition 2 By a *linear parametric function* we mean a scalar parameter of the form $\mathbf{a}^T \boldsymbol{\beta}$, where $\mathbf{a}$ is a $p \times 1$ vector of constants. That is

$$\mathbf{a}^T \boldsymbol{\beta} = a_1 \beta_1 + \cdots + a_p \beta_p.$$

Definition 3 A linear estimator is a scalar random variable of the form $\ell^T \mathbf{Y}$, where ℓ is an $n \times 1$ vector of constants. A linear estimator $\ell^T \mathbf{Y}$ is said to be a **Linear Unbiased Estimator** (LUE) of $\mathbf{a}^T \boldsymbol{\beta}$ if $\mathbb{E}(\ell^T \mathbf{Y}) = \mathbf{a}^T \boldsymbol{\beta}$, for all $\boldsymbol{\beta} \in \mathbb{R}^p$. The parameter $\mathbf{a}^T \boldsymbol{\beta}$ is said to be **estimable** if it has an LUE.

Lemma 1 $\mathbf{a}^T \boldsymbol{\beta}$ is estimable if and only if $\mathbf{a} \in \mathcal{C}(\mathbf{X}^T) = \mathcal{R}(\mathbf{X})$.

Proof : Estimability means there exists $\ell \in \mathbb{R}^n$ such that $\mathbb{E}(\ell^T \mathbf{Y}) = \mathbf{a}^T \boldsymbol{\beta}$, which holds if and only if $\ell^T \mathbf{X} \boldsymbol{\beta} = \mathbf{a}^T \boldsymbol{\beta}$ for all $\boldsymbol{\beta} \in \mathbb{R}^p$, if and only if $\ell^T \mathbf{X} = \mathbf{a}^T$.

- The lemma above shows that not all linear parametric functions may be estimable. It depends on $\mathcal{R}(\mathbf{X})$.
- If $\mathbf{X}$ has full column rank p , then all linear parametric functions $\mathbf{a}^T \boldsymbol{\beta}$, for any $\mathbf{a} \in \mathbb{R}^p$, are estimable.
- Conversely, if all linear parametric functions are estimable, then $\mathbf{X}$ must have full column rank.
- In multiple regression if all the covariates (i.e. the columns of $\mathbf{X}$) are linearly independent, all linear parametric functions are estimable. Such models are called *full rank models*.
- Linear models like one way ANOVA model with parameter vector $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_r)^T$, where r is the number of levels of the factor, do not have full rank. So only certain linear parametric functions are estimable.

Example 1 Consider the following model:

$$\mathbb{E}(Y_{ij}) = \alpha_i + \beta_j, \quad i = 1, 2; \quad j = 1, 2.$$

We can express the model in matrix form as

$$\mathbb{E} \begin{bmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \beta_1 \\ \beta_2 \end{bmatrix}.$$

$\mathbf{X}$ is the 4×4 matrix on the right-hand side.

- Define $S = \{(l_1, l_2, m_1, m_2) : l_1 + l_2 = m_1 + m_2\}$. Notice that S is a subspace of $\mathbb{R}^4$.
- Clearly for the i -th row of $\mathbf{X}$, $x_{i1} + x_{i2} = x_{i3} + x_{i4}$. So for any $z \in \mathcal{R}(\mathbf{X})$, $z \in S$. This implies that $\mathcal{R}(\mathbf{X}) \subseteq S$.
- Check that the dimension of S is 3, and also that $\text{rank}(\mathbf{X}) = 3$. This implies that $\mathcal{R}(\mathbf{X}) = S$. Thus $l_1 \alpha_1 + l_2 \alpha_2 + m_1 \beta_1 + m_2 \beta_2$ is estimable if and only if $l_1 + l_2 = m_1 + m_2$.

1.4 Best linear unbiased estimator

Definition 4 An LUE $\ell^T \mathbf{Y}$ of $\mathbf{a}^T \boldsymbol{\beta}$ is said to be a **Best Linear Unbiased Estimator (BLUE)** of $\mathbf{a}^T \boldsymbol{\beta}$ if

$$\text{Var}(\ell^T \mathbf{Y}) \leq \text{Var}(\mathbf{m}^T \mathbf{Y})$$

for any LUE $\mathbf{m}^T \mathbf{Y}$ of $\mathbf{a}^T \boldsymbol{\beta}$.

The following are useful properties of a BLUE.

- A BLUE is a minimum variance estimator among the class of all LUE.

- Every estimable linear parametric function has a unique BLUE.

Definition 5 A linear function $\mathbf{c}^T \mathbf{Y}$ is called a **Linear Zero Function (LZF)** if $\mathbb{E}(\mathbf{c}^T \mathbf{Y}) = 0$ for all $\boldsymbol{\beta} \in \mathbb{R}^p$. Clearly $\mathbf{c}^T \mathbf{Y}$ is a linear zero function if and only if $\mathbf{c}^T \mathbf{X} = 0$.

Lemma 2 An LUE $\ell^T \mathbf{Y}$ is a BLUE if and only if $\text{Cov}(\ell^T \mathbf{Y}, \mathbf{c}^T \mathbf{Y}) = 0$ for each LZF $\mathbf{c}^T \mathbf{Y}$.

Proof: If $\mathbf{m}^T \mathbf{Y}$ is a LUE of $\mathbf{a}^T \boldsymbol{\beta}$, then $\mathbf{m}^T \mathbf{Y} - \ell^T \mathbf{Y}$ is a linear zero function. Now,

$$\text{Var}(\mathbf{m}^T \mathbf{Y}) = \text{Var}(\mathbf{m}^T \mathbf{Y} - \ell^T \mathbf{Y} + \ell^T \mathbf{Y}) = \text{Var}(\mathbf{m}^T \mathbf{Y} - \ell^T \mathbf{Y}) + \text{Var}(\ell^T \mathbf{Y}) \geq \text{Var}(\ell^T \mathbf{Y}).$$

This proves the “if part” of the argument.

To prove the “only if part”, note that for any $\lambda \in \mathbb{R}$, $\ell^T \mathbf{Y} + \lambda \mathbf{c}^T \mathbf{Y}$ is an LUE of $\mathbf{a}^T \boldsymbol{\beta}$. Clearly,

$$\text{Var}(\ell^T \mathbf{Y} + \lambda \mathbf{c}^T \mathbf{Y}) \geq \text{Var}(\ell^T \mathbf{Y}) \Rightarrow \lambda^2 \text{Var}(\ell^T \mathbf{Y}) + 2\lambda \text{Cov}(\ell^T \mathbf{Y}, \mathbf{c}^T \mathbf{Y}) \geq 0.$$

The quadratic in λ on the right hand side has two distinct solutions, viz., $\lambda = 0$ and $\lambda = -2\text{Cov}(\ell^T \mathbf{Y}, \mathbf{c}^T \mathbf{Y})/\text{Var}(\ell^T \mathbf{Y})$, unless $\text{Cov}(\ell^T \mathbf{Y}, \mathbf{c}^T \mathbf{Y}) = 0$. Hence, unless $\text{Cov}(\ell^T \mathbf{Y}, \mathbf{c}^T \mathbf{Y}) = 0$, the above expression can take negative values, which is a contradiction.

The next result, which is referred to as the *Gauss-Markov Theorem* is key to linear estimation theory under Gauss-Markov set up. To state this, we first note that a least squares estimator of $\boldsymbol{\beta}$ is denoted by $\hat{\boldsymbol{\beta}}$ and is of the form $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$. The latter is unique only if $\mathbf{X}^T \mathbf{X}$ is non-singular. However, the “fitted value” $\mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{P}_X \mathbf{Y}$ is always unique.

Theorem 1 If $\mathbf{a}^T \boldsymbol{\beta}$ is estimable, then $\mathbf{a}^T \hat{\boldsymbol{\beta}}$ is its unique BLUE.

Proof : $\mathbf{a} \in \mathcal{R}(\mathbf{X})$ implies that $\mathbf{a} \in \mathcal{C}(\mathbf{X}^T) = \mathcal{C}(\mathbf{X}^T \mathbf{X})$, which implies that $\mathbf{a} = \mathbf{X}^T \mathbf{X} \mathbf{u}$ for some $\mathbf{u} \in \mathbb{R}^p$.

Uniqueness: Using the above,

$$\mathbf{a}^T \hat{\boldsymbol{\beta}} = \mathbf{u}^T (\mathbf{X}^T \mathbf{X}) \hat{\boldsymbol{\beta}} = \mathbf{u}^T \mathbf{X}^T \mathbf{Y}.$$

So $\mathbf{a}^T \hat{\boldsymbol{\beta}}$ is a unique for all possible values of $\hat{\boldsymbol{\beta}}$.

Unbiasedness: Taking expectation we get:

$$\mathbb{E}(\mathbf{a}^T \hat{\boldsymbol{\beta}}) = \mathbb{E}(\mathbf{u}^T \mathbf{X}^T \mathbf{Y}) = \mathbf{u}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\beta} = \mathbf{a}^T \boldsymbol{\beta}$$

for all $\boldsymbol{\beta} \in \mathbb{R}^p$.

Minimum variance : First, recall that if $\ell^T \mathbf{Y}$ is any LUE of $\mathbf{a}^T \boldsymbol{\beta}$, then $\ell^T \mathbf{X} = \mathbf{a}^T$.

Clearly, $\ell^T \mathbf{Y} - \mathbf{u}^T \mathbf{X}^T \mathbf{Y}$ is an LZF. We show that $\text{Cov}(\ell^T \mathbf{Y} - \mathbf{u}^T \mathbf{X}^T \mathbf{Y}, \mathbf{u}^T \mathbf{X}^T \mathbf{Y}) = 0$, from which we can conclude that $\mathbf{a}^T \hat{\boldsymbol{\beta}}$ is the BLUE by using Lemma 2, since $\ell^T \mathbf{Y}$ is an arbitrary LUE.

Notice that

$$\text{Cov}(\ell^T \mathbf{Y} - \mathbf{u}^T \mathbf{X}^T \mathbf{Y}, \mathbf{u}^T \mathbf{X}^T \mathbf{Y}) = (\ell^T - \mathbf{u}^T \mathbf{X}^T) \text{Var}(\mathbf{Y}) \mathbf{X} \mathbf{u} = \sigma^2 (\ell^T - \mathbf{u}^T \mathbf{X}^T) \mathbf{X} \mathbf{u}.$$

Further, since $\ell^T \mathbf{X} = \mathbf{a}^T$, it follows that

$$(\ell^T - \mathbf{u}^T \mathbf{X}^T) \mathbf{X} \mathbf{u} = (\ell^T \mathbf{X} - \mathbf{u}^T \mathbf{X}^T \mathbf{X}) \mathbf{u} = (\mathbf{a}^T - \mathbf{a}^T) \mathbf{u} = 0.$$

This completes the proof.